

IceApple Website ? Remediation Plan

Critical Issues

1. Root layout breaks because cn is used as a tagged template (src/app/layout.tsx).
 - Update body className to call cn(moskFont.variable, inter.variable, "font-main").
 - Add a simple render test to guarantee the layout mounts without throwing.
2. SendGrid setup runs at import time in api/client and api/career routes.
 - Move sgMail.setApiKey and env validations inside each handler; return 500 when missing.
 - Guard outbound email calls with try/except and log sanitized errors only once per request.
 - Document required SENDGRID_* variables in env.example.md and deployment checklist.
3. download-resume endpoint allows path traversal and serves every file as PDF.
 - Store original filename, MIME type, and generated key when saving uploads.
 - Sanitize requests by rejecting names that contain path separators or unmatched keys.
 - Stream the file using createReadStream and set Content-Type from the stored metadata.
 - Add automated test covering invalid file names and missing files.
4. Career upload overwrites filenames and forces .pdf extension.
 - Capture file.name when available, normalize it, and persist alongside the generated id.
 - Use the captured extension to build the storage key and SendGrid attachment metadata.
 - Expose the original name in the download endpoint response headers.

Quality Improvements

5. useDeviceType drives layout changes client side leading to hydration mismatch.
 - Replace deviceType checks with Tailwind responsive classes or CSS container queries.
 - Where runtime checks are unavoidable, gate them behind useEffect to keep SSR consistent.
 - Audit pages with React profiler after the change to confirm layout stability.
6. Navigation menu drops navigationMenuTriggerStyle styles.
 - Wrap navigationMenuTriggerStyle() in cn() so focus and keyboard styles are preserved.
 - Add a Storybook interaction test to ensure arrow-key navigation works on desktop.
7. Enquiry form lacks Content-Type header, loading state, and surfaced errors.
 - Send fetch with headers: { 'Content-Type': 'application/json' } and disable submit while pending.
 - Surface API failures in the UI (toast or inline alert) and re-enable the button afterwards.
 - Write a Playwright test submitting the form with mocked API responses (success + failure).
8. Key images render with empty alt text.
 - Provide descriptive alt strings in hero, enquiry, and map sections or mark them as decorative when appropriate.
 - Run axe-core against each page to confirm the accessibility regression is fixed.

Maintainability

9. Inconsistent import paths and skipped TypeScript safety.
 - Replace ../../.. imports with the configured @ alias using a codemod or eslint rule.
 - Disable allowJs and skipLibCheck once stray JS files are migrated, then fix exposed typing bugs.
10. Blogs page performs synchronous disk I/O and logs directly to console.
 - Switch to async read with fs.promises.readFile and memoize the parsed feed with cache().
 - Remove console.log or wrap it behind debug logging controlled by NODE_ENV.
 - Add an integration test confirming the page renders fallback content when XML parsing fails.
11. Missing automated test coverage for critical flows.
 - Add unit tests for API handlers using supertest and mocked MongoDB/SendGrid dependencies.
 - Create end-to-end smoke tests covering contact form submission, resume upload, and navigation.
 - Configure CI to run lint, type-check, unit, and e2e suites on every pull request.